

MPTP Activates c-Jun NH₂-Terminal Kinase (JNK) and Its Upstream Regulatory Kinase MKK4 in Nigrostriatal Neurons In Vivo

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Abstract: The neuropathology of Parkinson's disease is reflected in experimental animals treated with the selective nigrostriatal dopaminergic neurotoxin MPTP. Neurons exposed to MPTP (MPP⁺) express morphological features of apoptosis, although the intracellular pathways that produce this morphology have not been established. The c-Jun NH₂-terminal kinase (JNK) signaling cascade has been implicated as a mediator of MPTP-induced apoptotic neuronal death based on the ability of CEP-1347/KT-7515, an inhibitor of JNK activation, to attenuate MPTP-induced nigrostriatal dopaminergic degeneration. In these studies, MPTP-mediated activation of the JNK signaling pathway was assessed in the nigrostriatal system of MPTP-treated mice. MPTP elevated levels of phosphorylated JNK and JNK kinase (MKK4; also known as SEK1 or JNKK), by 2.5- and fivefold, respectively. Peak elevations occurred soon after administration of MPTP and coincided with peak CNS levels of MPP⁺. Increased MKK4 phosphorylation, but not JNK phosphorylation, was found in the striatum, suggesting that activation of MKK4 occurs in injured dopaminergic terminals. Both JNK and MKK4 phosphorylations were attenuated by pretreatment with *l*-deprenyl, indicating that these phosphorylation events were mediated by MPP⁺. Moreover, CEP-1347/KT-7515 inhibited MPTP-mediated MKK4 and JNK signaling at a dose that attenuates MPTP-induced dopaminergic loss. These data implicate this signaling pathway in MPTP-mediated nigrostriatal dopaminergic death and suggest that it may be activated in the degenerative process in Parkinson's disease. **Key Words:** MPTP—MPP⁺—c-Jun NH₂-terminal kinase—c-Jun NH₂-terminal kinase—Phosphorylated c-Jun NH₂-terminal kinase—Parkinson's disease—Apoptosis. *J. Neurochem.* **75**, 1200–1208 (2000).

1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) is a potent and selective nigrostriatal dopaminergic neurotoxin that produces many of the neuropathological features of idiopathic Parkinson's disease (PD) in humans, nonhuman primates, and mice (Heikkilä et al., 1984a,b; Agid, 1991; Langston, 1996; Blanchet et al., 1998). A key feature in MPTP-induced neurotoxicity is inhibition of complex I of the mitochondrial electron transport chain by its neuro-

toxic metabolite, 1-methyl-4-phenylpyridinium (MPP⁺) (Nicklas et al., 1985; Vyas et al., 1986; Saporito et al., 1992; Tipton and Singer, 1993). In idiopathic PD, complex I deficiencies have been identified in the substantia nigra of PD patients, and these deficiencies may be sufficient to precipitate neurodegeneration in the disease (Swerdlow et al., 1996; Schapira et al., 1998). The similarities between MPTP-induced neurotoxicity and idiopathic PD indicate that MPTP produces a mechanistically relevant model of PD. Thus, events secondary to MPP⁺ inhibition of mitochondrial respiration may be applicable to neurodegenerative events that occur in PD.

Programmed cell death mechanisms, secondary to complex I deficiencies, may play a role in the neurodegenerative processes in MPTP-induced toxicity and PD. MPTP (MPP⁺) produces morphological features of apoptosis, including nuclear chromatin condensation and membrane blebbing, as well as DNA laddering in PC12 cells, SH-SY5Y cells, cultured mesencephalic neurons, and cerebellar granule cells and in the substantia nigra of mice (Dipasquale et al., 1991; Hartley et al., 1994; Mochizuki et al., 1994; Tatton and Kish, 1997; Fall and Bennett, 1999). Several intraneuronal pathways of programmed cell death have been implicated in MPTP-induced neurotoxicity. Mice overexpressing Bcl-2 (an antiapoptotic protein) or deficient in p53 (a proapoptotic protein) are resistant to MPTP-induced neurotoxicity (Trimmer et al., 1996; Offen et al., 1998; Yang et al., 1998). It is interesting that nuclear chromatin condensation and an increase in nuclear translocated nuclear factor κ B were found in the substantia nigra from PD

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Abbreviations used: ERK, extracellular signal-regulated kinase; JNK, c-Jun NH₂-terminal kinase; MEKK1, mitogen-activated protein kinase kinase kinase 1; MKK4, c-Jun NH₂-terminal kinase kinase; MPP⁺, 1-methyl-4-phenylpyridinium; MPTP, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; PBS, phosphate-buffered saline; PD, Parkinson's disease; SAPK, stress-activated protein kinase.

patients, suggesting that programmed cell death mechanisms may participate in dopaminergic neurodegeneration in the disease (Anglade et al., 1997; Hunot et al., 1997).

One pathway of apoptotic death involves the activation of c-Jun NH₂-terminal kinase (JNK), a component of a stress-activated protein kinase (SAPK) pathway (Derijard et al., 1995; Ham et al., 1995; Xia et al., 1995; Eilers et al., 1998). In PC12 cells and sympathetic ganglia, inhibition of JNK activity by introduction of dominant-negative forms of JNK, the JNK substrate c-Jun, or the JNK activator JNK kinase (MKK4; also known as SEK1 or JNKK) block cell death evoked by trophic factor deprivation (Ham et al., 1995; Xia et al., 1995; Eilers et al., 1998). In primary cultures of motor neurons, the indolocarbazole CEP-1347/KT-7515 inhibits the activation of the JNK signaling pathway and promotes motor neuronal survival at similar low nanomolar concentrations (Maroney et al., 1998). In vivo, CEP-1347/KT-7515 attenuates MPTP-mediated nigrostriatal dopaminergic loss, indicating that the JNK signaling pathway may be activated with MPTP administration (Saporito et al., 1999).

The current studies were designed to extend these findings by investigating whether MPTP activated the JNK signaling pathway in the nigrostriatal pathway. In these studies, MPTP increased levels of phosphorylated JNK and its upstream regulatory kinase MKK4 in the nigrostriatal system. These increased phosphorylation events were inhibited by preadministration of CEP-1347/KT-7515. Results from these studies implicate this kinase signaling pathway in MPTP-mediated dopaminergic degeneration and suggest that this pathway may be active in degenerating dopaminergic neurons in PD.

MATERIALS AND METHODS

Animals, neurotoxin, and drug treatment

All procedures using animals were conducted in accordance with the NIH *Guide for the Care and Use of Laboratory Animals* and were approved by the Institutional Animals Care and Use Committee. Male C57 black mice (Charles River) weighing between 20 and 25 g (8–12 weeks of age) were used for all experiments. Animals were maintained on a standard 12-h light/dark cycle and were allowed food and water ad libitum.

MPTP (Research Biochemicals International, Natick, MA, U.S.A.) was dissolved in phosphate-buffered saline (PBS) at concentrations of 4 mg/ml. Mice received a single subcutaneous injection at a volume of 10 ml/kg to produce doses of 40 mg/kg, respectively. At the indicated times, mice were killed, their brains were removed, and striata and the midbrain containing the substantia nigra were dissected free and frozen at -80°C until ready for analysis. In the indicated studies, *l*-deprenyl was administered intraperitoneally 24 and 4 h before MPTP at a dose of 2.5 mg/kg per injection. CEP-1347/KT-7515 was administered subcutaneously at the indicated doses (0.1 and 1.0 mg/kg) 4 h before MPTP.

MPP⁺ brain level measurements

At 1, 3 and 5 h after a single MPTP injection, groups of mice were killed, the brains were removed, and the striata were

dissected free, frozen, and stored at -80°C . Striata were homogenized in 0.4 M perchloric acid (50 mg of tissue/ml) and then centrifuged at 15,000 *g* for 15 min. Concentrations of MPP⁺ were measured by HPLC with UV detection as previously described (Giovanni et al., 1991).

Immunoblots

Isolated striata or substantia nigra were homogenized (50 mg/ml) in lysis buffer (10 mM Tris, 50 mM NaCl, 0.03 μM sodium pyrophosphate, 50 mM sodium fluoride, and 1% Triton X-100) containing 1 mM phenylmethylsulfonyl fluoride, 20 $\mu\text{g}/\text{ml}$ aprotinin, and 1 mM sodium vanadate. After homogenization, samples were centrifuged at 15,000 *g* for 15 min. The supernatant was saved and analyzed for protein concentration (BCA: Pierce Laboratories, Rockford, IL, U.S.A.). Ten micrograms of protein (per sample) was separated on a 10–20% Tris-glycine polyacrylamide gel and transferred to nitrocellulose membranes (0.45 μm). The blot was incubated with antibody and, subsequently, in anti-mouse (or antirabbit) alkaline phosphatase conjugate. The ECF detection system (Amersham) was used to visualize protein following the directions of the manufacturer. Blots were incubated with ECF reagent for 10 min, dried, and scanned on a PhosphorImager (Molecular Dynamics). Protein levels were quantified by measuring the density of each band on the PhosphorImager.

For the characterization study of the phospho-MKK4 antibody, 100 μg of baculovirus-expressed recombinant human MKK4 (from Dr. Sherri Meyer) was incubated in the presence of 50 μg of mitogen-activated protein kinase kinase 1 (MEKK1; Upstate Biotechnology) for 30 min at 30°C . Phosphorylated and nonphosphorylated MKK4 (1 μg of protein) was resolved on a polyacrylamide gel, transferred to nitrocellulose, and probed with both the phospho- and total MKK4 antibodies as described.

All primary antibodies were used at a 1:1,000 dilution. Antibodies used were the polyclonal phosphospecific MKK4 (Thr 223, Ab 9151; New England Biolabs, Beverly, MA, U.S.A.), monoclonal, phosphospecific SAPK/JNK (Thr 183, Tyr185, Ab 9255; New England Biolabs), monoclonal JNK antibody that detects JNK 1 and 2 (catalogue no. 15691A; Pharmingen), and polyclonal MKK4 (SC-834; Santa Cruz Biotechnology).

Immunohistochemistry

A free-floating method for immunohistochemical detection of phospho-MKK4 was used in these studies. After treatment with MPTP (as described above) mice were transcardially perfused with 10 ml of cold PBS followed by 25 ml of 4% paraformaldehyde in PBS. Brains were postfixed in 4% paraformaldehyde at 4°C . Following postfixation, brains were saturated in 30% sucrose in PBS for cryopreservation. Fixed frozen midbrains were sectioned coronally at 30 μm thickness on a sliding microtome. The phospho-MKK4 antibody (New England Biolabs) was used at a 1:2,500 dilution, and sections were incubated for 48 h. After washing in PBS, sections were incubated with biotinylated anti-rabbit IgG fraction for 1 h at a 1:100 dilution in PBS. A standard avidin–biotin complex method (Vectastain Elite Kit; Vector Laboratories, Burlingame, CA, U.S.A.) was used for antibody visualization. Sections were incubated for an additional 30 min with an avidin-biotinylated horseradish peroxidase complex followed by subsequent incubation with 3,3'-diaminobenzidine (1 mg/ml) and 0.03% H₂O₂.

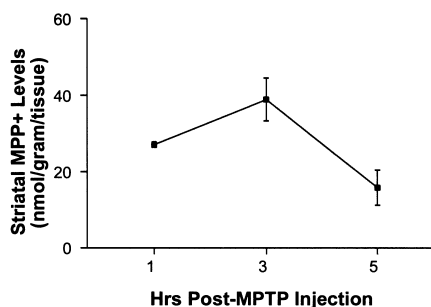


FIG. 1. MPP⁺ striatal levels. MPTP was administered at a single 40 mg/kg, s.c., dose, and striatal MPP⁺ levels were measured 1, 3, and 5 h after dosing. Data are mean $\pm$ SEM (bars) values ($n = 6$ per time point).

Statistical analysis

All results expressed as mean $\pm$ SEM values. Data were analyzed by ANOVA, and differences between treatments were determined by post hoc Newman-Keuls test unless otherwise indicated. Means were considered statistically significant at $p < 0.05$.

RESULTS

MPP⁺ brain levels

MPP⁺ striatal levels were assessed 1, 3, and 5 h after a single subcutaneous 40 mg/kg dose of MPTP (Fig. 1). Levels of MPP⁺ were measurable at all time points after administration. Peak levels of 40 nmol/g of tissue were found 3 h after administration. These levels were significantly lower 5 h after injection. These data are consistent with previously published reports of MPP⁺ levels in the striata (Giovanni et al., 1991).

Antibody characterization

The antibodies for these studies were used according to the specifications of the manufacturer. In brain homogenates, the JNK and phospho-JNK antibodies detected two protein bands of 46 and 54 kDa (Fig. 2). These molecular size bands correspond to the reported sizes of cloned and expressed JNK (Gupta et al., 1996). The phospho-specific JNK antibody detected protein bands of the same size in UV-treated 293 cells but not in control cells (data not shown). This antibody has been previously characterized in sympathetic neurons (Anderson and Tolkovsky, 1999). The MKK4 antibodies detected a single protein band of 43 kDa in brain homogenates (Fig. 3). This molecular size corresponds to the reported size of MKK4 (Derijard et al., 1995). In characterization experiments, we generated phosphorylated MKK4 by preincubating recombinant MKK4 in the presence of MEKK1. The phosphospecific MKK4 antibody detected recombinant prephosphorylated MKK4 but not nonphosphorylated MKK4 at 1 μ g of protein. These data are shown in Fig. 3A. Both the JNK and MKK4 (total kinase) antibodies have been previously characterized (Flood et al., 1998).

Time course of MPTP-induced JNK and MKK4 phosphorylation

Substantia nigra. In these studies, mice were treated with a 40 mg/kg dose of MPTP. This dose of MPTP produces complete loss of striatal dopaminergic terminal markers and substantial loss of tyrosine hydroxylase-immunoreactive neurons in the substantia nigra (Saporito et al., 1999). MPTP administration increased the levels of p46 and p54 phospho-JNK proteins by 1.5- and three-fold, respectively, as compared with PBS-treated controls (Fig. 2). Peak elevations for both isoforms occurred 4 h after administration of MPTP. By 24 h, phospho-JNK had returned to control levels. These blots were stripped and reprobed with total JNK antibody, which detects both phospho- and non-phospho-JNK. There was no corresponding change in levels of either protein band identified by this JNK antibody.

An identical time course of MPTP administration was used to assess levels of phospho-MKK4 in the substantia nigra. MPTP administration increased levels of phospho-

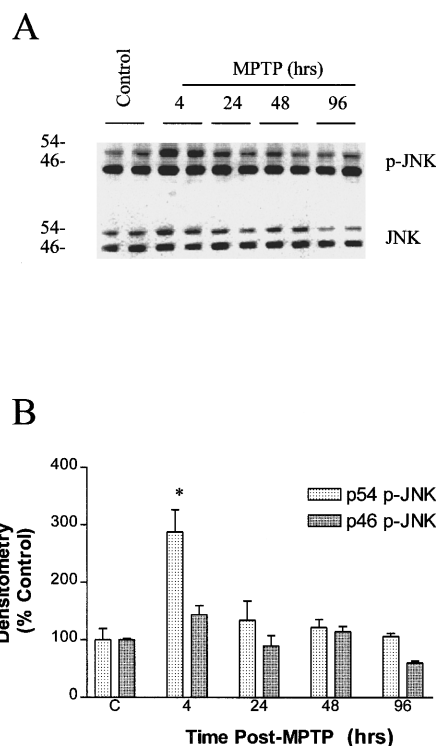


FIG. 2. MPTP increases phospho-JNK (p-JNK) levels in mid-brain homogenates. **A:** MPTP was administered at a single dose of 40 mg/kg, s.c., and the midbrain (containing the substantia nigra) was dissected at the indicated times after MPTP administration, homogenized, electrophoresed on a polyacrylamide gel, and transferred to a nitrocellulose midbrain. A representative immunoblot of MPTP-mediated increases in phospho-specific JNK is shown. **B:** Graphical representation of MPTP-mediated increases in p-JNK levels. Control animals were treated subcutaneously with PBS, and levels of p-JNK were determined 4 h after administration. Data are from two experiments and are average $\pm$ SEM (bars) values ($n = 6$ per group). * $p < 0.05$, statistical difference from control (PBS-treated) animals.

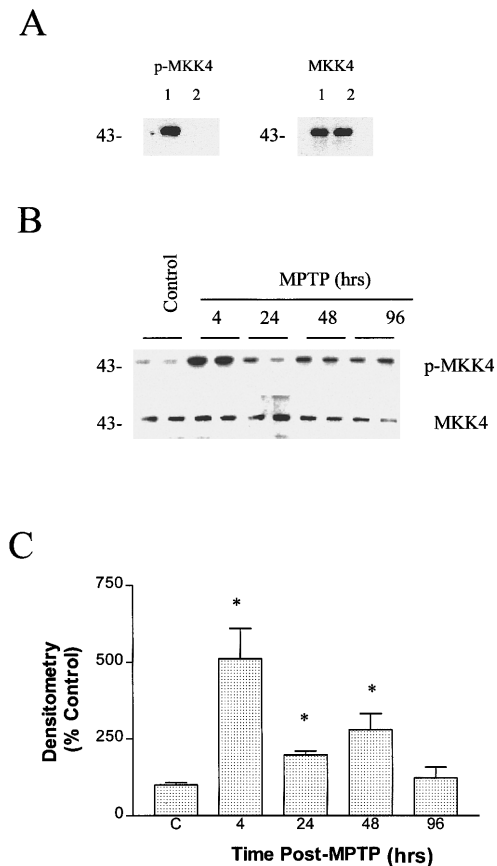


FIG. 3. MPTP increases phospho-MKK4 (p-MKK4) levels in mid-brain homogenates. **A:** Characterization of the p-MKK4 antibody. Lanes 1, recombinant MKK4 (1 μ g) prephosphorylated with MEKK1; lanes 2, recombinant MKK4 (1 μ g), nonphosphorylated. Blots were probed with either p-MKK4 antibody or total MKK4 antibody as indicated. **B:** MPTP was administered at a single dose of 40 mg/kg, s.c., and the midbrain (containing the substantia nigra) was dissected at the indicated times after MPTP administration, homogenized, electrophoresed on a polyacrylamide gel, and transferred to a nitrocellulose membrane. A representative immunoblot of MPTP-mediated increases in phospho-specific MKK4 is shown. **C:** Graphical representation of MPTP-mediated increases in p-MKK4 levels. Control animals were treated subcutaneously with PBS, and levels of p-MKK4 were determined 4 h after administration. Data are from two experiments and are average $\pm$ SEM (bars) values ($n = 6$ per group). * $p < 0.05$, statistical difference from control (PBS-treated) animals.

MKK4 in the substantia nigra 4 h after injection (Fig. 3). Maximal activation was sixfold over control levels. Two-fold increases were apparent 24 h after MPTP administration, and these same twofold increases were sustained for up to 48 h postdosing. At 96 h postdosing there was not a difference between MPTP-treated and control tissues. These blots were stripped and reprobed with an antibody that detected total MKK4 levels (non-phospho- and phospho-MKK4). There was no corresponding change in levels of total MKK4 using this antibody (data not shown).

Striata. There was no detectable change in levels of phospho-JNK (or non-phospho-JNK) in the striata of MPTP-treated mice at the same time points (and the same animals) in which we examined phosphorylation in the substantia nigra (Fig. 4A). However, MPTP-mediated increases in phospho-MKK4 levels were found in striatal homogenates (Fig. 4A and B). The magnitude of induction and time course of induction were similar to those found in the substantia nigra samples. Induced levels of phospho-MKK4 were detectable until 96 h after MPTP administration. These blots were stripped and reprobed with antibody that detected total MKK4 levels. MPTP did not affect levels of total MKK4 (data not shown).

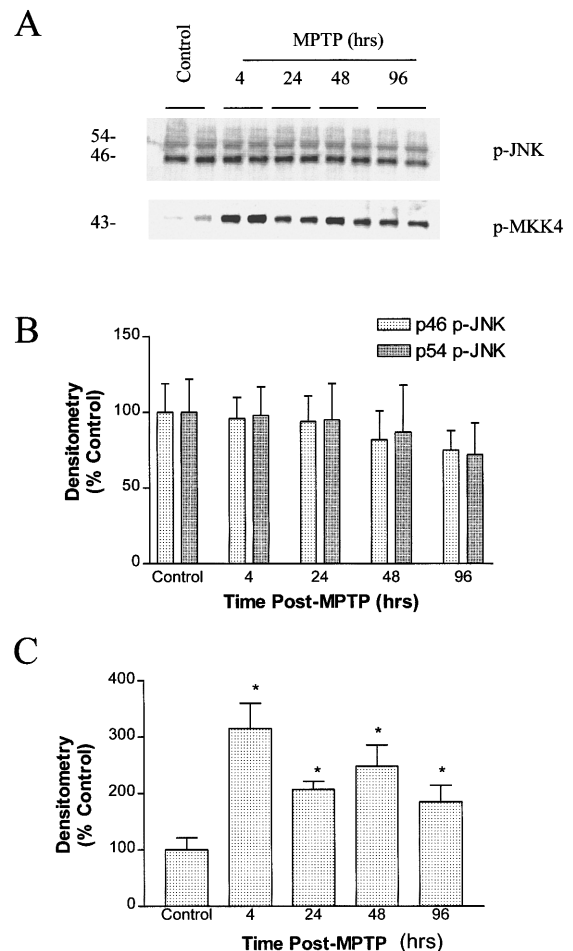


FIG. 4. MPTP-increases phospho-MKK4 (p-MKK4) but not phospho-JNK (p-JNK) in striatal homogenates. Striata were removed from the same animals in which substantia nigra was removed. Levels of p-JNK and p-MKK4 as well as their non-phosphorylated counterparts were analyzed in these striatal homogenates. **A:** Representative immunoblot of p-MKK4 and p-JNK in the striatum. **B:** Graphical representation of lack of p-JNK induction in the striatum. **C:** Graphical representation of MPTP-mediated increases in p-MKK4 levels. Data are from two experiments and are average $\pm$ SEM (bars) values ($n = 6$ per group). * $p < 0.05$, statistical difference from control (PBS-treated) animals.

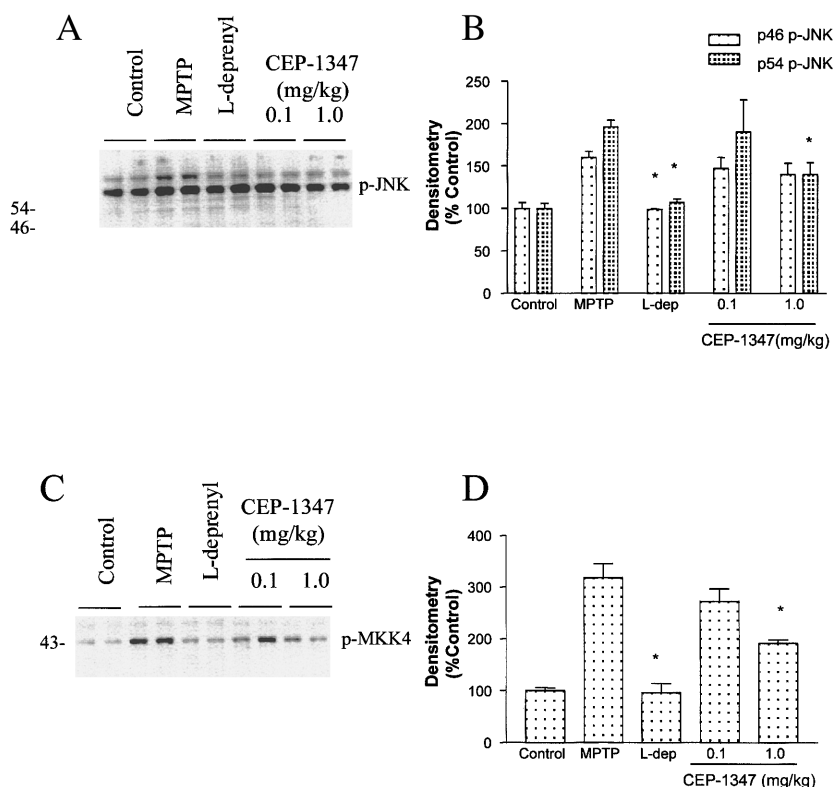


FIG. 5. *L*-Deprenyl (*L*-dep) and CEP-1347/KT-7515 inhibit MPTP-mediated increases in phospho-MKK4 (p-MKK4) and phospho-JNK (p-JNK) levels in the substantia nigra. *L*-Dep (2.5 mg/kg per intraperitoneal injection) was preadministered to mice 18 and 2 h before MPTP administration. CEP-1347 (0.1 and 1.0 mg/kg) was administered 4 h before MPTP administration. Midbrain (containing the substantia nigra) was removed 4 h after MPTP administration and assessed for (A) p-JNK immunoblot, (B) p-JNK graphical representation, (C) p-MKK4 immunoblot, and (D) p-MKK4 graphical representation. For the graphs, data are from two experiments and are average $\pm$ SEM (bars) values ($n = 5$ per group). * $p < 0.05$, statistical difference from MPTP-treated vehicle control.

Inhibition of MPTP-mediated MKK4 and JNK phosphorylation by *L*-deprenyl and CEP-1347/KT-7515

Preadministration of the monoamine oxidase inhibitor *L*-deprenyl (2.5 mg/kg dose, i.p.) completely attenuated the MPTP-mediated increase in phospho-JNK and phospho-MKK4 levels in midbrain homogenates (Fig. 5). The dose of *L*-deprenyl used in these studies completely inhibits formation of MPP⁺ (through inhibition of monoamine oxidase-B) in the CNS and blocks MPTP-induced dopaminergic neurotoxicity (Heikkilä et al., 1984b).

The JNK signaling pathway inhibitor CEP-1347/KT-7515 was assessed for its ability to inhibit MPTP-mediated MKK4 and JNK phosphorylation. CEP-1347/KT-7515 was assessed at doses of 0.1 and 1 mg/kg s.c. This dose of CEP-1347/KT-7515 attenuates the MPTP-mediated loss of tyrosine hydroxylase-immunoreactive neurons in the substantia nigra and excitotoxic injury of the nucleus basalis magnocellularis (Saporito et al., 1998, 1999). A single 4-h preadministration of CEP-1347/KT-7515, at a dose of 1 mg/kg, partially attenuated the MPTP-mediated increase in phospho-JNK and phospho-MKK4 levels without affecting levels of total MKK4 or JNK protein (Fig. 5). A lower dose (0.1 mg/kg) was not sufficient to inhibit phospho-MKK4 or phospho-JNK induction.

Immunohistochemistry of MPTP-mediated increases in phospho-MKK4

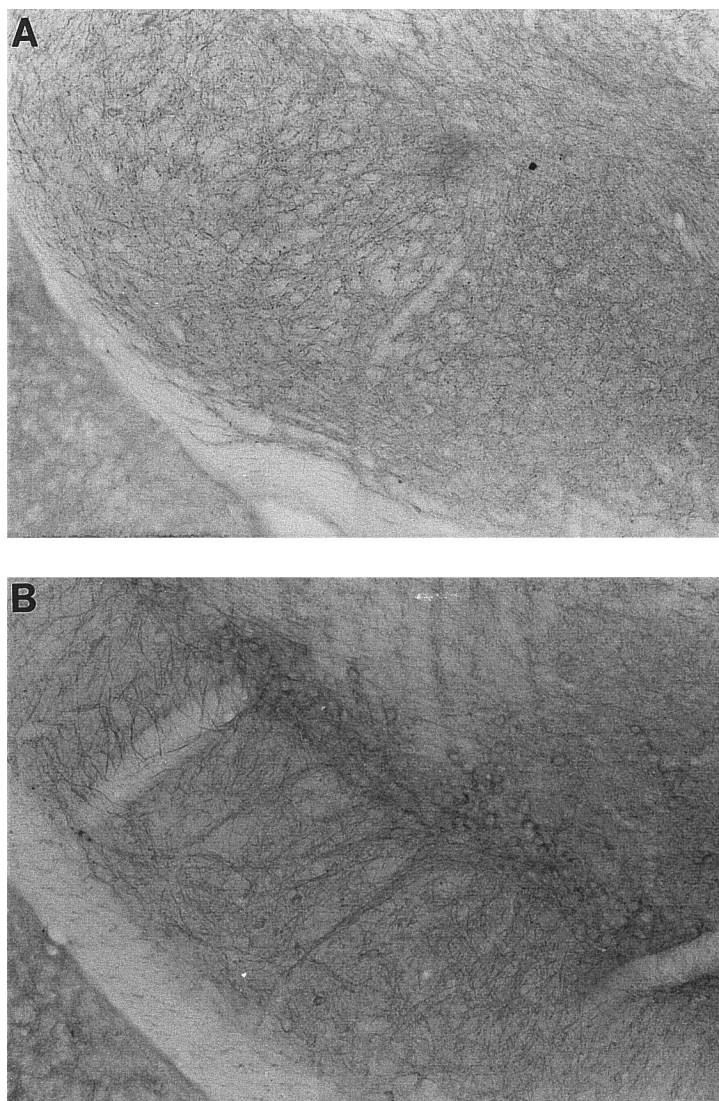
MPTP-mediated increases in phospho-MKK4 levels were also assessed by immunohistochemistry. Phospho-

MKK4 was at very low levels in the substantia nigra of control mice. MPTP administration increased staining of phospho-MKK4 in the region of the substantia nigra. This increased phospho-MKK4 staining could be eliminated by removing the primary antibody from the protocol or by preincubating the tissue sections with protein phosphatase I. In the substantia nigra pars compacta, phospho-MKK4 was found in the cell bodies and corresponding neurites that extended to the substantia nigra pars reticulata (Fig. 6). In addition, diffuse staining was apparent in cell bodies and neurites of the substantia nigra pars reticulata. Phospho-MKK4 staining was apparent in the cytoplasm and was not detected in the nucleus. Staining appeared to be restricted to neurons; no staining was apparent in glial or nonneuronal cells.

DISCUSSION

MPTP administration to experimental animals produces a pattern of neurodegeneration and a neuropathology that is similar to that seen in the brains of PD patients (Heikkilä et al., 1984a; Agid, 1991; Langston, 1996). These similarities suggest that MPTP may be useful in identifying intracellular pathways of neurodegeneration that are relevant to neurodegenerative events that occur in PD. In the present studies, MPTP administration to mice increased levels of phosphorylated MKK4 and its downstream substrate, JNK, in the substantia nigra and increased the levels of phosphorylated MKK4 in the striatum. These elevated levels of phosphokinases occurred shortly after administration of MPTP and well

FIG. 6. Phospho-MKK4 immunohistochemistry in (A) PBS- and (B) MPTP-treated animals. MPTP or PBS was administered subcutaneously, and animals were perfused with 4% paraformaldehyde 4 h after MPTP administration. Brains were sectioned at 30 μ m thickness and immunostained with phospho-MKK4 antibody. MPTP increases phospho-MKK4 levels in the cytoplasm and neurites of substantia nigra pars compacta and pars reticulata neurons. Phospho-MKK4 staining is absent when either the primary or secondary antibody is excluded in the immunohistochemistry procedure.



before loss of nigrostriatal dopaminergic neurons. These phosphorylation events were likely mediated by MPP^+ as the monoamine oxidase-B inhibitor *l*-deprenyl attenuated the MPTP response. MPTP did not alter the levels of total JNK or MKK4, indicating that the changes represented increases in the phosphorylation state of the protein and not a change in the levels of the corresponding nonphosphorylated form of the kinase.

There was a close temporal correlation between formation of MPP^+ (within 4 h) and activation of MKK4 and JNK in the CNS. MPP^+ inhibition of mitochondrial respiration occurs rapidly after exposure of cells to MPP^+ and concomitantly with MPP^+ formation in the CNS (Rollemma et al., 1990; Chan et al., 1991; Cassarino et al., 1997). These data suggest that the MPTP-mediated MKK4 and JNK activation was closely coupled to MPP^+ inhibition of mitochondrial respiration in the striatum and substantia nigra. MPP^+ inhibition of mitochondrial respiration produces several rapid secondary events, in-

cluding depletions in intracellular ATP and increased levels of markers of oxidative stress (Chan et al., 1991; Cassarino et al., 1997). Of these events, increases in oxidative stress may be an important mediator of events secondary to MPP^+ inhibition of mitochondrial respiration. For example, in isolated mitochondria MPP^+ promotes opening of the mitochondrial transition pore and release of apoptotic mediators, including cytochrome c, via an oxidative mechanism (Cassarino et al., 1999). Oxidative stress is also an important activator of the JNK signaling cascade. In differentiated PC12 cells and cultured sympathetic neurons, increases in oxidative stress induced by antisense inactivation of superoxide dismutase increase JNK activity (Maroney et al., 1999). Moreover, increases in intracellular oxidative stress induced by exposure of striatal neurons to toxic levels of dopamine activate the MKK4/JNK signaling cascade (Luo et al., 1998). Thus, the MPP^+ -mediated increases in oxidative stress are a potential mechanism linking inhi-

bition of mitochondrial respiration to activation of MKK4 and JNK. Alternatively, increases in intracellular calcium levels have been found to be an important activator of mitogen-activated protein kinases. In PC12 cells, increases in intracellular calcium levels induced by mitochondrial inhibition activate extracellular signal-regulated kinases (ERKs) types 1 and 2 (Luo et al., 1997). However, JNK appears to be much less sensitive than the ERKs to elevated intracellular calcium levels (Schwarzschild et al., 1999). Further studies in this area are needed to define the sequence of events that couple MPP⁺ inhibition of mitochondrial respiration to activation of MKK4 and JNK.

Activated MKK4 and JNK are considered to be mediators of some forms of stress-induced cell death. In PC12 cells, inactivation of JNK with dominant-negative mutants attenuates death of PC12 cells deprived of trophic support or in HeLa cells exposed to an oxidative stress (Xia et al., 1995; Wang et al., 1998). Moreover, dominant-negative forms of MKK4 block cell death (a) in sympathetic ganglia transfected with MEKK1, (b) in PC12 cells exposed to manganese, and (c) in striatal cultures exposed to toxic doses of dopamine (Eilers et al., 1998; Hirata et al., 1998; Luo et al., 1998). In vivo, activated JNK has been implicated as a mediator of dopaminergic neuronal death in a developmental model of target deprivation-induced apoptosis (Oo et al., 1999). Moreover, mice deficient in JNK3 or that have mutations in c-Jun phosphorylation sites are resistant to kainic acid-induced seizures and neuronal death in the CA3 region of the hippocampus (Yang et al., 1997; Behrens et al., 1999). However, there is also evidence correlating JNK activation to neuronal regeneration and survival. In transected dorsal root ganglion, JNK activity was sustained until reinnervation of the target tissue, suggesting that JNK can participate in the regenerative response (Kenney and Kocsis, 1998). Furthermore, embryonic mice deficient in both JNK1 and JNK2 (but not JNK3) had increased levels of apoptotic neurons in the developing forebrain and decreased apoptotic neurons in the developing hindbrain, indicating that these kinases are important in promoting both neuronal survival and programmed cell death during development (Kuan et al., 1999).

The early MPTP-mediated activation of MKK4 and JNK appears to be more closely associated with neuronal death than with a failed attempt at promoting neuronal survival and regeneration. For example, the early activation of MKK4 and JNK coincides with depletions in ATP and mitochondrial failure and precedes loss of any dopaminergic neuronal markers or loss of neuronal integrity (Rollema et al., 1990; Chan et al., 1991; Jackson-Lewis et al., 1995). This rapid and transient increase in MKK4 and JNK levels is consistent with this process triggering a programmed cell death mechanism. In contrast, ischemia, medial forebrain bundle axotomy, or transection of the sciatic nerve also produces increases in JNK activity; however, these increases occur well after initial insult and parallel the time course of neuronal loss

in the substantia nigra or during the time course of regeneration in dorsal root ganglion (Herdegen et al., 1998; Kenney and Kocsis, 1998). Furthermore, MPTP-mediated MKK4 and JNK activations were inhibited with preadministration of CEP-1347/KT-7515 at doses that attenuate MPTP-induced neurodegeneration (Saporito et al., 1999). The similarities between the neuroprotective dose and the dose needed to inhibit MKK4 and JNK activation suggest that this signaling pathway is coupled to MPTP-mediated neurodegeneration. It is interesting that mice deficient in p53 are resistant to MPTP-mediated neurotoxicity (Trimmer et al., 1996). p53 is a proapoptotic nuclear transcription factor that is phosphorylated and activated by JNK (Fuchs et al., 1998). These data indicate that a direct link exists between activation of the JNK pathway and MPTP-mediated dopaminergic neuronal death.

The MPTP-mediated activation of phospho-MKK4 and phospho-JNK is likely restricted to dopaminergic neurons of the nigrostriatal dopaminergic system. In support of this statement, MKK4 and JNK mRNA and protein are found in high concentrations in neurons (including neurons of the substantia nigra) with little found in glia or other nonneuronal tissue (Carletti et al., 1995; Carboni et al., 1997; Flood et al., 1998). Moreover, MPP⁺ is known to be accumulated and to selectively damage dopaminergic neurons of the nigrostriatal pathway without directly affecting other neuronal types or nonneuronal cells (Heikkila et al., 1984a; Tipton and Singer, 1993).

Data from the current studies also indicate that MKK4 activation, but not JNK activation, occurs within dopaminergic terminals in the striatum. In support of this concept, immunoblot analysis showed that MPTP-mediated increases in phospho-MKK4 occurred in striatal homogenates. Because striata are enriched in dopaminergic terminals and do not contain dopaminergic cell bodies and MPP⁺ is accumulated within dopaminergic terminals, these data suggest that MKK4 activation occurs within these terminals. In the striata, activation of JNK is unlikely to occur at time points earlier than 4 h because this time point is sufficiently early to see JNK activation in the substantia nigra. Moreover, MKK4 is an upstream regulatory kinase of JNKs, suggesting that if there was JNK activation in this region, it should occur simultaneously with MKK4 activation. The immunohistochemistry results show that MPTP-mediated increases in phospho-MKK4 occur in neurites (as well as cell bodies) of the substantia nigra pars compacta and pars reticulata. The MPTP-mediated increase in phospho-MKK4 in dopaminergic terminals in the absence of increased levels of phospho-JNK indicates that MKK4 may have alternative downstream substrates to JNK in this brain region. JNK activation is associated with nuclear transcriptional events, and, unlike MKK4, JNK1 translocates to the nucleus during ischemia and reperfusion in perfused rat heart (Mizukami et al., 1997). Thus, the lack of JNK activation in the striatum (which only contains dopami-

nergic terminals) is consistent with the association of JNK with nuclear-mediated events.

MPTP-mediated dopaminergic neurotoxicity provides a useful system to study neurodegenerative events that may occur in PD. The biochemical sequelae of events that occur between MPP⁺ inhibition of mitochondrial respiration and dopaminergic degeneration are beginning to be described and may be useful in identifying neurodegenerative pathways in the disease. Using the MPTP model, events leading to apoptosis have been implicated as mediators of neuronal death. The present studies demonstrate that MPTP mediates the phosphorylation of MKK4 and JNK in vivo. Because these kinases are involved in stress-induced apoptotic death in several systems, the current data implicate this kinase signaling pathway in MPTP-induced dopaminergic degeneration. These data also suggest that this kinase signaling pathway may mediate some of the neurodegenerative mechanisms that occur in PD.

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